

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2013-14

Zoology

Paper: ZOB-603 — Immunology, Microbiology, Environmental Biology and Biotechniques

Time: Three hours

Full Marks: 70

Note: Answer two questions from each Section including Question No. 1 of each Section which is compulsory. Use a separate answer book for each Section. The figures in the right-hand margin indicate marks.

SECTION – A
(Immunology) (Marks: 17½)

Note: Answer two questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

(a) Define the following: [3 × 1 = 3]

(i) Immunogenicity

(ii) Proteosomes

(iii) Epitopes.

(b) Indicate whether the following statement is True or False. Explain by giving justification in 2–3 sentences: [2]

(i) IgM antibody is the most abundant and can cross the placenta.

(c) Fill in the blank: [½]

(i) Cell mediated immunity is brought about by _____ cells.

Question 2. Write short notes on the following: [3 × 4 = 12]

(a) Complement system

(b) Antigen processing and presentation

(c) Primary and secondary lymphoid organs.

Question 3. How does B cell activate and secrete the antibodies? [12]

SECTION – B
(Microbiology) (Marks: 17½)

Note: Answer two questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

(a) Define the following: [5 × 1 = 5]

- (i) Attenuated micro-organism
- (ii) Proteobacteria
- (iii) Virus
- (iv) Opportunistic pathogen
- (v) Biopolymer.

(b) Fill in the blanks: [½]

- (i) *Clostridium* produces _____ or _____ which breaks the collagen network and causes gas gangrene.

Question 2. Write short notes on the following: [3 × 4 = 12]

- (a) Biopesticides
- (b) Antimicrobial drug susceptibility testing
- (c) Biodegradation of xenobiotics.

Question 3. Answer the following parts:

(a) Differentiate between the following: [2 × 4 = 8]

- (i) Cytolytic toxins and Enterotoxins
- (ii) Archaea and Bacteria.

(b) Discuss the importance of intestinal microflora. [4]

SECTION – C
(Environmental Biology) (Marks: $17\frac{1}{2}$)

Note: Answer two questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

(a) State whether the following statement is True or False, giving reasons in not more than 2–3 sentences: [2]

(i) The average number of new individuals produced by a population in a given time is called biotic potential.

(b) Define the following: [3 × 1 = 3]

(i) Food web

(ii) Biotic community

(iii) Threatened species.

(c) Fill in the blank: [$\frac{1}{2}$]

(i) The total amount of photosynthetic energy captured by autotrophs in a given period of time is called _____.

Question 2. Answer the following parts:

(a) Describe the energy flow in a forest ecosystem. [6]

(b) Give an account of mineral conservation. [6]

Question 3. Write short notes on the following: [3 × 4 = 12]

(a) Population characteristics

(b) Water conservation

(c) Benefits of biodiversity.

SECTION – D
(Biotechniques) (Marks: 17½)

Note: Answer two questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False. Justify your answer in brief:
[2 × 1½ = 3]
- (i) Chromatographic separations are based on wavelength.
 - (ii) UV-visible spectrophotometer is routinely used in the quantitative determination of biological macromolecules.
- (b) Define numerical aperture of a microscope. [1]
- (c) Expand the following: [3 × ½ = 1½]
- (i) RCF
 - (ii) PAGE
 - (iii) ICC.

Question 2. Answer the following parts:

- (a) Describe the principle and application of fluorescence microscope. [4 + 2 = 6]
- (b) What is tissue fixation? Write down the procedure of preparation of a stained histological slide. [2 + 4 = 6]

Question 3. Answer the following parts:

- (a) What do you mean by culture media? State its importance for different types of animal cell culture. [2 + 4 = 6]
- (b) Write short notes on the following: [2 × 3 = 6]
- (i) Cryopreservation
 - (ii) pH meter.

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